

MeshEcho: Confidence-Aware Route Admission for LoRa Mesh IoT

Gao Zu Quan Zhi
Shenzhen University, Shenzhen, China

Abstract—LoRa mesh IoT networks trade source-confirmed reliability against scarce airtime. Flooding adds redundancy but creates collisions, whereas a cached source route can select a weak multi-hop path. We present MeshEcho, a confidence-aware route-admission policy with bounded recovery, and evaluate it with a reproducible packet-level simulator. Version 2.1.22 uses matched candidate exposure, independent reception streams, a distinct min-hop baseline, and explicit generalization cases. In the calibrated 50-node, 8.25-km matrix over 20 seeds, MeshEcho reaches 0.720 ACK-confirmed completion at 28.1 s airtime, versus 0.502/38.5 s for matched source routing and 0.661/27.6 s for ETX and fixed-SF ETT. The paired MeshEcho minus source-route ACK-PDR difference is +0.218 (95% CI [+0.041,+0.395]); the destination-PDR interval includes zero. Unconditioned random pairs and a 100-node, 3–5-hop case retain an ACK-PDR advantage over ETX, but at a small airtime cost. Temporal block fading with short route TTL moves MeshEcho to 0.324 ACK PDR and 71.3 s airtime, so route aging is a boundary diagnostic rather than a universal claim. The evidence supports a bounded source-confirmed-reliability–airtime operating point in simulation, not topology-independent dominance or hardware validity.

Index Terms—LoRa mesh, Internet of Things, route admission, packet delivery ratio, airtime

I. Introduction

LoRa is attractive for low-power IoT because it combines long link budget with inexpensive radios [1]. A star topology is insufficient for obstructed or infrastructure-poor deployments, motivating multi-hop extensions [2]–[4]. Multi-hop forwarding is expensive: a low data rate occupies the channel for a long time, while a flood multiplies collision opportunities. A cached source route reduces redundancy, but one weak hop can invalidate an otherwise useful path.

MeshEcho records lightweight link-quality evidence during discovery, ranks candidate paths by confidence, and invokes bounded recovery only when a route is uncertain. The contribution is not a new link metric; it is an explicit policy that couples confidence-based admission, a matched discovery budget, and bounded recovery, then tests that policy against shortest-candidate, ETX, ETT, and flooding baselines. The implementation is a simulation prototype, not a production mesh stack.

This paper makes four contributions:

- a reproducible packet-level LoRa mesh harness with time-on-air, propagation, shadowing, half-duplex collision/capture, route discovery, and energy accounting;

- a per-seed quality-gated, non-saturated ICC matrix with matched candidate exposure and paired confidence intervals;
- MeshEcho-only component attribution, a distinct min-hop baseline, and random-pair, deep-multihop, and temporal-fading generalization evidence; and
- an explicit boundary separating source-confirmed completion from destination delivery, cache-aging diagnostics, and simulation-only validity.

II. Related Work

Prior ICC LoRa studies include analytical and simulation work on coverage, spreading-factor allocation, and frame error rates [5]–[9], as well as testbed validation [10], [11]. Multihop studies expose the flooding–airtime tradeoff [2], [4] and show that routing state and packet timing matter [12], [13]. ETX is a standard expected-transmission metric [14]; our fixed-SF ETT row is a sanity baseline because constant packet time-on-air makes it numerically coincide with ETX. We therefore ask a narrower question: does confidence admission change the reliability–airtime operating point under one shared packet model?

III. Protocol and Model

A. Network model

Nodes are uniformly placed in a square and share one half-duplex LoRa channel. Unless noted, packets have 32-byte payloads, 17 dBm transmit power, 125-kHz bandwidth, coding rate 4/5, capture threshold 6 dB, and SF7. Airtime follows the standard LoRa symbol calculation [15]. Log-distance path loss with log-normal shadowing produces SNR; reception is a logistic function of SNR margin, and overlapping packets are accepted only when the desired signal passes capture. The simulator records collisions, route-cache events, discovery success, and configured TX/RX energy.

B. Routing policies

Managed flooding rebroadcasts with duplicate suppression. Matched source-routing floods a request, collects all destination replies arriving in a 2-s window, and selects the shortest returned candidate. ETX and ETT use the same discovery and forwarding machinery but minimize

$$C_{\text{ETX}} = \sum_i p_i^{-1}, \quad C_{\text{ETT}} = \sum_i \text{ToA}_i p_i^{-1},$$

TABLE I
Calibrated ICC parameters.

Nodes / area	50 / 8250 m × 8250 m
Duration / load	600 s / 1 flow min ⁻¹
Radio	32 B, 17 dBm, 125 kHz, CR 4/5, SF7
Path loss / shadowing	$n = 2.75$ / $\sigma = 4$ dB
Discovery / route TTL	2 s / 600 s
Pairs / graph	24 / PRR ≥ 0.90 , 2–4 hops
Seeds / quality gate	20 / mean hops ≥ 2 , below-0.99 PRR $\geq 10\%$

where p_i is the observed per-hop PRR. The min-hop baseline ignores link quality after equal candidate exposure. MeshEcho selects the candidate with the highest confidence, using path length only as a tie-break; a 0.025 per-hop penalty and a linear age penalty (maximum 0.1) are applied to the confidence score. If no usable route is admitted, a bounded two-hop fallback is allowed. The ICC runs disable adaptive timeout retries. The historical Smart-CALM firmware/simulator namespace remains in the repository but is not an ICC protocol, ablation, or conclusion.

IV. Experimental Design

A. Primary matrix and metrics

The primary matrix has 50 nodes in an 8250 m square, 600-s runs, mixed traffic at one flow/min, SF7, 4-dB shadowing, and 20 seeds. For each seed, 24 source–destination pairs are selected from a shared analytical graph with edge PRR ≥ 0.90 and graph distance 2–4 hops. The quality gate requires a complete pair pool, mean graph distance at least 2, and at least 10% of direct links below 0.99 PRR. In this calibrated graph every selected pair is exactly two hops and all primary route-cache-hit counts are zero. Thus the headline estimand is first-discovery route admission, not general cache reuse. All protocols share node placement, traffic, shadowing, reception streams, discovery window, and route TTL. Reported intervals are two-sided 95% paired CIs computed from seed-level differences.

B. Generalization strata

Three separate 20-seed strata avoid pooling different estimands. Random pairs use 50 nodes in an 18-km square with mixed traffic at 4 flows/min. Deep multihop uses 100 nodes in a 21-km square, unicast at 2 flows/min, and 24 pairs selected at graph distance 3–5; the 21-km calibration was chosen before the final run because the 20-km setting failed the complete-pool gate for seed 12. Temporal-fading cases reuse four connected pairs in the 8.25-km square with 6-dB block fading every 60 s, comparing 600-s and 30-s route TTLs. These cases report limits and are not pooled with Table II.

V. Results

A. Primary and mechanism evidence

MeshEcho improves over matched source routing by +0.218 ACK PDR (95% CI [+0.041, +0.395]) and reduces airtime by 10.3 s ([−12.6, −8.1]). Destination PDR

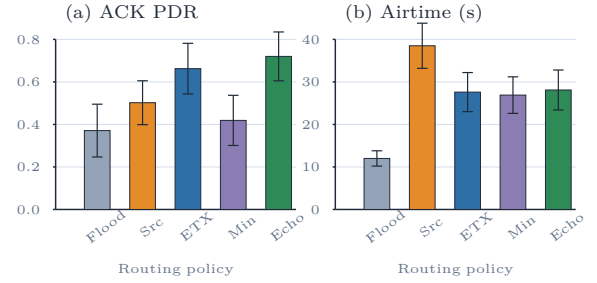


Fig. 1. Primary calibrated matrix. Bars are seed-level means and whiskers are two-sided 95% confidence intervals. The left-to-right policy order is managed flooding, matched source-route (Src), ETX/ETT, min-hop (Min), and MeshEcho (Echo). ETX and fixed-SF ETT coincide under the shared packet-time model and are shown as one metric bar.

changes by +0.103 ([−0.086, +0.292]), so the strongest claim concerns source-confirmed completion under the simulator’s reverse-path ACK model. MeshEcho versus ETX/ETT is +0.059 ACK PDR ([−0.108, +0.226]), and versus min-hop is +0.301 ([+0.161, +0.441]); the latter is a distinct shortest-path baseline, not an oracle.

The no-confidence row is an end-to-end policy-bundle comparison: discovery success, candidate count, and route length also change. The controlled seven-node route-conflict audit supplies narrower evidence. With a 7.5-dB weakened short branch, both candidates appear in 90% of seeds; MeshEcho selects the longer branch in every observed conflict and improves ACK PDR by $+0.221 \pm 0.096$ over shortest-candidate selection. This supports candidate admission in that controlled topology, not a universal confidence advantage.

B. Generalization and temporal fading

Random pairs retain an ACK-PDR advantage over ETX while removing fixed-pair reuse; the difference is accompanied by a small airtime increase. The deeper case shows the same direction at 3–5 hops, but MeshEcho is not the lowest-airtime policy. Under temporal fading, shortening the TTL increases repair and airtime for every routing policy and reduces MeshEcho to 0.324 ACK PDR. The fading results therefore define a stale-route boundary rather than proving that confidence aging solves time-varying channels.

C. Sensitivity and stress cases

At SF8, MeshEcho reaches 0.405 ACK PDR/61.7 s, while managed flooding reaches 0.434/39.2 s. At four flows/min, MeshEcho reaches 0.660/111.0 s versus 0.450/107.6 s for ETX and 0.409/50.5 s for flooding. These are operating points, not universal rankings. The 18-km, 3-seed quality probe passes its pair and link gates but has only 0.178 MeshCore-like discovery success; it is reported as a route-discovery stress diagnostic, not a neutral data-plane comparison.

TABLE II

Primary calibrated matrix (20 seeds). Values are means; $\pm$ denotes the two-sided 95% CI half-width. ACK PDR requires a reverse-path ACK, while destination PDR does not.

Protocol	ACK PDR	Dest. PDR	Airtime (s)	Discovery	Multi-hop route	Mean hops
Managed flooding	.371 $\pm$.124	1.000 $\pm$.000	12.0 $\pm$ 1.8	n/a	n/a	n/a
Matched source-route	.502 $\pm$.103	.636 $\pm$.110	38.5 $\pm$ 5.3	.720	.803	1.82
ETX metric	.661 $\pm$.119	.741 $\pm$.098	27.6 $\pm$ 4.6	.861	.843	1.91
Fixed-SF ETT metric	.661 $\pm$.119	.741 $\pm$.098	27.6 $\pm$ 4.6	.861	.843	1.91
Min-hop	.419 $\pm$.118	.435 $\pm$.113	26.9 $\pm$ 4.3	.637	.716	1.67
MeshEcho	.720 $\pm$.115	.739 $\pm$.115	28.1 $\pm$ 4.7	.891	.989	2.03

TABLE III

MeshEcho component attribution in the primary matrix. Deltas are MeshEcho minus the named variant; intervals are paired 95% CIs.

Variant	Δ ACK	95% CI	Δ airtime
No confidence	+.148	[+.052, +.244]	+4 s
No fallback	+.000	[+.000, +.000]	+1 s
No hop penalty	+.050	[−.029, +.130]	−0.2 s
No age penalty	+.000	[+.000, +.000]	+0.0 s

TABLE IV

MeshEcho ACK-PDR/airtime means in separate generalization strata.

Case	MeshEcho	ETX	Δ ACK	Δ airtime
Random pairs	.404/99.5	.308/98.1	+.096	+1.3 s
3–5-hop, 100 nodes	.696/160.4	.531/156.6	+.165	+3.9 s
Fading, TTL 600 s	.526/21.2	.450/20.2	+.076	+0.9 s
Fading, TTL 30 s	.324/71.3	.256/63.3	+.067	+8.0 s

D. Fairness and metric boundaries

The matched source-route baseline collects every destination reply arriving within the same discovery window as MeshEcho, so the primary contrast is not an immediate-reply versus delayed-reply artifact. The protocols nevertheless retain different forwarding and recovery semantics by design. We therefore report both ACK PDR and destination PDR: an ACK confirms source-side completion, whereas destination PDR can remain high when the reverse path fails. Airtime includes discovery, data, ACK, and recovery transmissions, so the reported operating point does not hide control cost.

The cache table is intentionally not a replacement for the primary matrix: it reuses four pairs and measures route-cache reuse under fading. Likewise, the deep case was calibrated to satisfy a complete pair pool before the final 20-seed run; its 21-km setting is disclosed rather than presented as an untouched robustness point. These choices make the estimand explicit and allow future work to replace pair conditioning, static shadowing, or the reverse-path ACK model independently.

VI. Discussion and Limitations

The evidence supports a bounded claim: in the calibrated shared harness, MeshEcho improves the source-confirmed reliability–airtime operating point over matched source routing, while ETX/ETT remain statistically in-

TABLE V

Secondary cache-aging diagnostic (20 seeds).

TTL	ACK PDR	Dest. PDR	Airtime (s)	Repairs
600 s	.786	.806	25.7	0.25
30 s	.472	.490	88.5	18.5

distinguishable in the primary comparison. Managed flooding remains the low-airtime reference, and the min-hop/deep/random strata show that the confidence policy is not an automatic dominance result. Cache reuse and temporal fading are secondary diagnostics: the 600-s versus 30-s cache experiment moves MeshEcho from 0.786/25.7 s to 0.472/88.5 s, with route repairs rising from 0.25 to 18.5 per run. These values are not pooled with the zero-hit primary estimand.

All results are packet-level simulation. The model includes path loss, shadowing, collision timing, capture, route discovery, and energy accounting, but omits waveform interference, oscillator offset, modem implementation timing, duty-cycle enforcement, variable route headers, and antenna effects. The confidence estimate and forwarding channel share the same realization in static runs; temporal block fading is a deterministic stress model, not hardware validation. A hardware or hardware-in-the-loop study remains necessary for radio-specific external validity.

A. Threats to interpretation

The quality gate is deliberately a measurement contract, not a guarantee of deployment realism. Pair selection from an analytical PRR graph removes degenerate disconnected samples, but it conditions the primary estimate on links that are usable in the static model. The random-pair stratum removes that conditioning at the cost of lower route completion, and the deep stratum tests a different scale and hop distribution. We report all three rather than averaging them into a single ranking.

The paired design shares topology, traffic, and reception streams, but it is not event-by-event common-random-number coupling: protocols transmit different numbers of packets and therefore encounter different collision histories. Confidence intervals are computed from independent seed-level differences, not from individual packets or flows. This avoids overstating precision while preserving the

TABLE VI
Version 2.1.22 evidence strata and estimands.

Artifact	Main perturbation	Estimand
Calibrated	PRR-qualified, 2-hop pairs	First discovery
Random pairs	Unconditioned source/destination	Generalization
Deep multihop	100 nodes, 3–5 hops	Scale/hop depth
Fading	6-dB blocks, TTL 600/30 s	Staleness boundary
Cache	Four repeated pairs	Cache reuse/aging

topology pairing. The no-confidence ablation also changes discovery success and route length, so it is a policy-bundle contrast, not a fixed-candidate causal estimate. Finally, the route-conflict audit has only seven nodes and is intentionally surgical; it cannot replace a random-topology comparison.

These boundaries determine what a future hardware study must preserve: the same pair schedule, discovery window, source-confirmed and destination metrics, airtime accounting, and route-TTL perturbation. A small SX126x or SX127x testbed with RSSI/SNR logs would test whether the simulated confidence ordering survives oscillator, antenna, modem-timing, and duty-cycle effects.

VII. Reproducibility

Version 2.1.22 releases source, tests, raw seed-level CSVs, summary CSVs, and Markdown reports. The main workflow is `OUT_PREFIX=meshecho_v2_1_22_icc2027 bash tools/run_icc_experiments.sh`. The sensitivity, generalization, cache, and route-conflict commands are versioned in `tools/run_icc_sensitivity_experiments.py`, `tools/run_icc_generalization_experiment.py`, `tools/run_icc_cache_experiment.py`, and `tools/run_route_conflict_experiment.py`. No human participants or personal data were involved; there are no known conflicts of interest or external funding.

A. Evidence manifest

The released artifacts keep raw observations separate from interpretation. The primary and sensitivity CSVs contain one row per protocol and seed; the generalization runner writes one file per estimand; and the Markdown reports record link quantiles, pair-selection rules, route-hop diagnostics, discovery success, and paired comparisons. The versioned component file contains only the MeshEcho policy variants, so a historical adaptive-firmware result cannot silently enter the ICC table. The quality gate is run before a connected multihop CSV is accepted, and its failure messages identify the seed and the violated condition. This separation makes it possible to reproduce the headline matrix while independently auditing selection, cache reuse, and temporal fading.

B. Audit checklist

Before interpretation, the release checks the protocol registry, the candidate-exposure window, the route-TTL

setting, and the independent channel stream. The calibrated and sensitivity runners reject incomplete connected pair pools before writing reports. The generalization runner records the selection rule for every stratum and reruns all protocols on the same seed and topology. The temporal-fading implementation derives its perturbation from the unordered link, seed, and time block, so protocol event ordering cannot change the fading realization. The cache diagnostic reports both route-cache hits and route repairs; the route-conflict audit records the selected path and both candidate confidence values. These checks are useful because a high aggregate PDR can otherwise hide a mostly one-hop or route-discovery-limited experiment.

The raw CSV schema also preserves negative and boundary cases. Managed flooding can have high destination PDR but low ACK PDR; ETX and ETT can coincide under fixed SF; min-hop can select a shorter but less reliable candidate; and MeshEcho can spend more airtime when confidence admission chooses a longer path. We retain these rows in the released summaries rather than replacing them with a single rank score. The paper’s numerical claims are generated from the Markdown reports, while the CSVs remain the authority for seed-level recomputation and alternative confidence intervals.

VIII. Conclusion

MeshEcho reaches 0.720 ACK PDR at 28.1 s in the calibrated multi-hop simulation, compared with 0.502/38.5 s for matched source routing and 0.661/27.6 s for ETX/ETT. Random-pair and deeper-hop cases preserve the direction of the ACK-PDR result with a small airtime cost, while temporal fading and short TTLs expose a clear stale-route boundary. Component and route-conflict evidence implicate confidence-aware candidate admission as the main effect in this workload. Version 2.1.22 therefore supports a reproducible, bounded MeshEcho simulation claim, not topology-independent dominance or physical validation.

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